

LeviBowl: Acoustic Levitation System for Interactive Handling of Large Lightweight Objects

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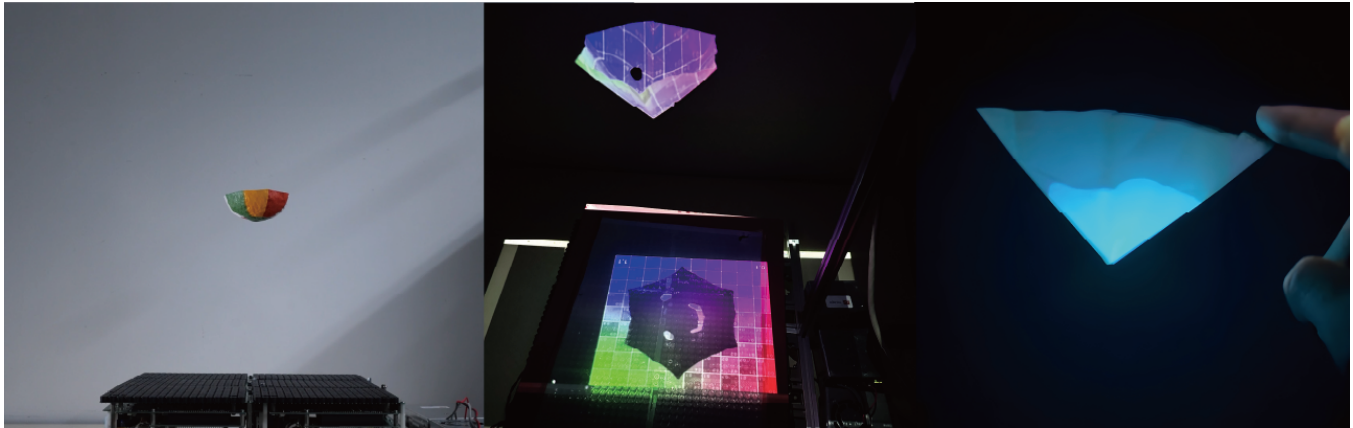


Figure 1: Objects that will be levitated by our method and demonstrations that will be presented.

Abstract

The application of acoustic levitation as a three-dimensional interface has been widely proposed. In this paper, we propose an interactive real-space three-dimensional interface using the technique we developed to interactively levitate objects larger than a wavelength farther than the wavelength. The proposed method can interactively control an object about 9.8 cm in size with simple controls. The proposed method can levitate and control an object sufficiently larger than its wavelength, and its workspace is as wide as $20 \times 20 \times 40$ cm, making it possible to realize an interface that is highly visible from a distance. In addition, since the shape of the levitated object is relatively robust in the proposed method, it is expected to have a wide range of applications beyond data visualization, such as allowing users to change the shape of the levitated object according to their specific needs and project images accordingly. This demonstration provides an interactive experience of levitation control.

CCS Concepts

• Human-centered computing → Interaction devices.

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1 Introduction

There is a technique called acoustic levitation, which uses acoustic radiation force, a nonlinear phenomenon of ultrasound, to levitate lightweight objects and visualize data in three dimensions. For example, Omirou et al. proposed a device to visualize three-dimensional pathways using levitated beads[7]. Gao et al. refer to a comprehensive data visualization system using acoustic levitation, including MixedReality and airborne ultrasound haptic technology[4]. Acoustic levitation methods that have been used to visualize data by acoustic levitation have mainly used standing waves. In the standing wave approach, an object can be stably levitated by trapping it in a node of sound pressure of a standing wave generated by a phased array or other means. However, the standing wave method limits the levitable object to half a wavelength or less. Currently widely used ultrasonic phased arrays use 40 kHz ultrasonic waves, so lightweight objects up to 5 mm in half wavelength can be levitated. This paper proposes an acoustic levitation method that interactively levitates objects larger than a wavelength sufficiently farther than the wavelength. Specifically, a

hemispherical object with a diameter of about 9.8cm can be levitated with a workspace of about $20\text{cm} \times 20\text{cm} \times 40\text{cm}$ by using a circular orbit single focus sound field. In terms of visualization of three-dimensional data as shown in previous studies, this method can realize an interactive three-dimensional information space with good visibility even from a distance because of its large range of motion and the size of the levitating object. The large size of the object also makes projection mapping possible, and it is expected to be applied to a wide variety of three-dimensional information representation.

2 Background

2.1 Acoustic levitation as an interface

With the recent development of ultrasonic phased array technology, many applications utilizing acoustic levitation of ultrasonic waves have been proposed. The range of applications is wide, including non-contact manipulation of chemicals[2][9], levitation observation of relatively small animals such as insects[13], and non-contact assembly of micro-objects[1]. Examples of interfaces proposed in the past include a display using a large number of levitated beads and a projector[6][5], a display using fast-moving particle trajectories[8][3], a multimodal interface using ultrasonic tactile sensation in the air, and a taste presentation using levitated edible droplets[11]. These are methods that achieve acoustic levitation mainly by creating standing waves and trapping objects with their sound pressure nodes[12]. In principle, this technique has the limitation that the size of the levitated object must be less than half a wavelength.

2.2 Interactive acoustic levitation of objects larger than a wavelength into the distance

As mentioned earlier, most previous interactive acoustic levitation has been mainly for objects smaller than a wavelength. No technique has been proposed to interactively levitate objects larger than a wavelength sufficiently farther than the wavelength. In this paper, we generate a time-averaged bowl-shaped acoustic radiation pressure field using a single-focus sound field moving in a circular orbit, and levitate an object about 10 times larger than the wavelength at a distance of several tens of wavelengths by placing the object on the bowl-shaped field.

A schematic diagram of the proposed method is shown in Fig. 2. The phase of the transducer is controlled to form a focal point at a single specified point. The acoustic radiation pressure creates a repulsive pressure field around the focal point. When this focus is scanned in a circular orbit to form a bowl-shaped sound field, the hemispherical levitating object placed on top of it will gain restoring force against perturbations. Since the phase of a single-focus sound field can be easily determined using the distance between the transducer and the focal point, the desired pressure field can be formed with less computation time than when forming complex sound fields that require optimization. When the center of this circular orbit is shifted continuously, a restoring force acts against the displacement and the levitated object follows the moving center of the circular orbit. This is used to interactively control the position of the levitated object. If the target is a polygonal pyramid, it is

also possible to rotate it around the vertical axis by changing the position of the discretized focal point.

The proposed method uses a phased array (created using six AUTD3 units [10]) formed by $384\text{mm} \times 454\text{mm}$ and 1494 transducers placed on a table facing upwards. We have found that a hemisphere or hexagonal pyramid with a base of 10cm and a height of 5cm, 0.608g, made of paraffin paper, is levitated within a workspace of about $20\text{cm} \times 20\text{cm} \times 40\text{cm}$. It can also move horizontally at a maximum speed of 9.8 cm/s and vertically at 23.0 cm/s, and can rotate around a vertical axis at a rate of approximately 1.0 rad/s. A schematic diagram of the experimental apparatus is shown in Fig. 2.

The conditions for the more specific control methods and levitation limits of the proposed method are being prepared for submission as a separate paper.

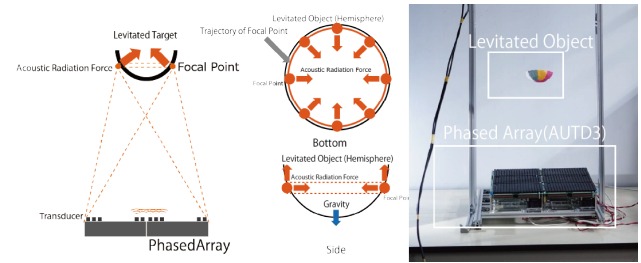


Figure 2: Left: Schematic of the proposed method, Right: Schematic of the equipment of the proposed method

3 Demonstration: LeviBowl

In this demonstration, visitors can experience levitation and control of lightweight objects and interactive information presentation through projection mapping. Specifically, various lightweight objects made of paraffin paper can be levitated in the air, and their position and rotation can be controlled using a controller. In addition, by projecting images on the paraffin paper using a projector placed at the top, visitors can experience interaction with the images projected in the air.

In the proposed method, an hexagonal pyramid made of paraffin paper with a radius of 10 cm, height of 6 cm, and weight of about 0.35 g is used as a levitation target on a sound pressure field with a circular orbital single focus. The position and angle of the levitation target can be controlled from any program by sending position information through the OSC protocol to an posture control program in Python. In our demonstration, the basic part of the novelty of the proposed system, “levitation control of an object larger than a wavelength at a distance sufficiently larger than the wavelength,” will be experienced to facilitate understanding of the intrinsic value of the system. In particular, we envision that the experience will be focused on the following functions.

3.1 Visual Programming Environment

As an application of the proposed method, we developed a system to control the motion of objects by block-based visual programming. This allows even non-programming specialists and small children

to easily control the position and posture of levitated objects. This programming system can be connected to a game controller or other input device to control it manually. While there are many robots that can move in a two-dimensional plane for similar purposes, the proposed robot can add a three-dimensional component of elevation movement to them. An example of a demonstration using visual programming is shown in Fig. 3. The person experiencing the demonstration can run a simple program using the visual programming environment we have implemented and experience the actual operation of the levitated object. In the actual demonstration, a motion control demonstration will be performed based on this visual programming environment.

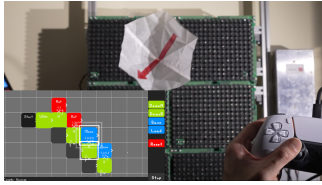


Figure 3: Demonstration of a visual programming environment and a controller created with the environment to operate a levitation object by receiving input from the controller.

3.2 Easily craftable levitation object

The proposed method is relatively robust to changes in the shape of the levitation target. Therefore, objects of various shapes created by simple crafting can be levitated. As long as the basic shape is a hexagonal pyramid and does not lose its balance, it is possible to make holes in it, attach pieces by cutting and pasting paraffin paper, or draw pictures with markers. Fig. 4 show figures of several levitating objects. People can levitate objects they create freely in the air. By combining this feature with the Visual Programming mentioned above, it has a wide range of applications. In our demonstration, several levitating objects are prepared and the participants can experience levitation with their own hands.

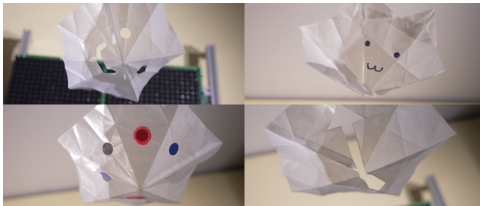


Figure 4: Various crafted objects are levitated. Upper left: object cut out to look like a face, upper right: object with ears and tail decorations attached and a face drawn with markers, lower left: object with colored markers attached for tracking, lower right: object cut out to look like an arrow to show the direction.

3.3 Information presentation using projection mapping

In the proposed method, a projector is attached to the top of the system and projects onto the levitated object from above, to superimpose visual information onto it. For example, arrows indicating direction, liquid in a cone, etc. can be projected. Since the size of the levitated object directly becomes the projectable surface, a relatively wide projection surface of about 10 cm in diameter can be obtained. In addition, since the levitated object is a pyramid, the projected content can be viewed from any direction in 360 degrees. A schematic diagram of the equipment for projection mapping is shown in Fig. 5. In the demonstration, visitors can observe projection mapping that works in sync with the levitated object and experience an interactive experience in which the projected liquid surface shakes when they touch the levitated object.

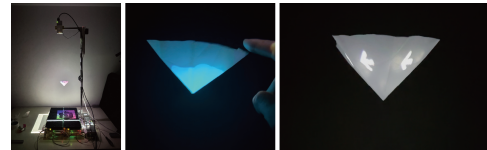


Figure 5: Left: Experimental equipment for projecting an object onto a levitated object. A small projector is placed directly above the object. Center: An example of demonstration. A liquid image is projected on a pyramid, and the surface of the liquid appears to sway when the object is touched. Right: An example of demonstration. An arrow moves in sync with the direction of rotation of the levitating object.

4 Further Applications

The demonstration proposed in this paper is a simple demonstration of ultrasonic control of the movement and rotation of an object and its application. In the future, we believe that the advantages of this method can be used to develop the following applications.

4.1 Educational Robots

The proposed method can make a levitated object operate in three dimensions through visual programming. By using this visual programming environment and the control system of the levitated object, beginning programmers and children can learn programming effectively. The object that can be levitated by the proposed method is about 10 cm in diameter, which is large enough to be observed in a relatively large space, such as a classroom.

4.2 Generation of storytelling and shadowgraph

The proposed method makes it possible to give the levitated objects some kind of characteristic by changing their shape through cutting and pasting. The proposed method could be used to extend storytelling techniques such as puppetry with levitation techniques. In addition, the proposed method also makes it possible to generate shadows of arbitrary shapes by cutting out the shape of the object to be projected. Using this technique, it is believed that more diverse storytelling, including shadow pictures, can be realized. Center part

of Fig. 1 shows how a shadowgraph is generated in combination with a projector.

4.3 Rehabilitations

The proposed method can move a 10-cm object in an arbitrary trajectory with a velocity of about 10 cm/s in three dimensions. By using the levitated object, it is believed that it will be possible to conduct training using the object as an indicator, and to conduct training to improve the physical functions of the elderly and for rehabilitation. For example, training that expands the range of motion of the arm is effective in improving frozen shoulder, and patients who have suffered a stroke often need physical rehabilitation to improve the physical paralysis that remains.

Previous methods have been proposed to improve patients' motivation for rehabilitation using VR technology, for example, but the proposed method may be able to realize similar rehabilitation intuitively from a display, without devices such as VR goggles.

4.4 Realization of large and extensive interfaces

The proposed method can levitate objects using a phased array in a single direction. This allows, for example, a phased array to be mounted on a robotic cart to obtain a large workspace. By using a phased array mounted on a robotic cart, for example, it is possible to create an interface that can follow the user and present information such as guidance in three dimensions.

5 Conclusion

In this paper, we propose a demonstration of an interactive levitating object using a technique that makes an object larger than the wavelength levitate at a distance greater than the wavelength, as well as a demonstration that combines the projector. The demonstration showed how to control the movement and posture with a controller and how to realize an interactive experience in combination with projection. The relatively large, open space at the top allows for movement control in a large workspace, and we consider this technology to be widely applicable as a fundamental technology for new interactions in the future.

Acknowledgments

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